

The empirical recurrence for OEIS A233975: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A233975 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 343$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233975 is “Number of $(n+1) \times (2+1)$ 0..6 arrays with every 2×2 subblock having the sum of the absolute values of all six edge and diagonal differences equal to 13.”. It has offset 1 and begins

1308, 8688, 59440, 453988, 3530540, 29109544, 242057156, 2076801776, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:07:01, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 343$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 343$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 686$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	11	c_{17}	55888647078051	c_{33}	−6144882855064084192	c_{49}	−745732926501390640
c_2	218	c_{18}	−205412655353490	c_{34}	−12574807632474555090	c_{50}	59891400187625904
c_3	−2763	c_{19}	−434223964913230	c_{35}	12386045956587720219	c_{51}	153387579650359712
c_4	−18299	c_{20}	1988224860702160	c_{36}	16202784522529513192	c_{52}	−18329885361139808
c_5	294278	c_{21}	2426276466444437	c_{37}	−18816464261473754511	c_{53}	−22005432377063312
c_6	711132	c_{22}	−14687045433277213	c_{38}	−15980114028246051853	c_{54}	3108361893602368
c_7	−17920404	c_{23}	−8970606818218775	c_{39}	21813315679261433140	c_{55}	2099322534108352
c_8	−6762661	c_{24}	83871142521860535	c_{40}	11830426338304097459	c_{56}	−305213487527680
c_9	705239900	c_{25}	14427826021121647	c_{41}	−19364336259445032127	c_{57}	−123843835293952
c_{10}	−593123053	c_{26}	−373127447224725510	c_{42}	−6343086187946366246	c_{58}	15999192697856
c_{11}	−19142505019	c_{27}	58951133990165657	c_{43}	13135915499046459493	c_{59}	3989693159424
c_{12}	32296499743	c_{28}	1298656357735298144	c_{44}	2273297819672350748	c_{60}	−347750834176
c_{13}	372295030173	c_{29}	−541175393032359862	c_{45}	−6763845362579532769	c_{61}	−53250785280
c_{14}	−879638723215	c_{30}	−3541166713791662988	c_{46}	−408410915846254680	c_{62}	922091520
c_{15}	−5303661975817	c_{31}	2227325458092028974	c_{47}	2614021160567708124		
c_{16}	15853808593545	c_{32}	7557014833232297310	c_{48}	−56828782016844040		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

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